Oussama Essamadi

CV

A software engineer in Tangier, writing software since 2016. I build full-stack platforms in Java, Spring Boot and Angular for Intesa Sanpaolo, serverless systems on AWS, and AI platforms built on retrieval and agents. Technical senior specialist at NTT Data, and a PhD candidate in artificial intelligence.

Whereabouts

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Experience

Technical senior specialist, NTT Data

Tangier · April 2025 to now

Full-stack engineer and stream lead for Intesa Sanpaolo. On **Documentum MSID**, a document management platform built from scratch, I deliver it end to end: Java 17 Spring Boot microservices over REST and GraphQL, an Angular 19 front end, fine-grained access control, versioned storage and search on Elasticsearch. Around it, I improve the existing microservices and the shared Angular libraries, and write the JasperReports templates for invoices and regulatory statements. Java 21, 17 and 8, Spring Boot, Angular, Maven and GitLab CI/CD.

Software engineer, NTT Data

Tangier · December 2022 to March 2025

For **Intesa Sanpaolo**, I took over the commercial stream's Anage portal and led its move from Angular 4 and Java 6 to Angular 7 and Java 8, clearing critical security issues and bringing the build up to date, from Maven to Gradle. On the CJHUB suite I led the accessibility work that brought it to WCAG 2.1 AA, and rolled out a new design system across every one of its apps, with the design, QA and operations teams, without downtime.

On **xNGAS**, a serverless platform on AWS in Python: legacy code refactored and better tested, new microservices, infrastructure as code with the CDK, CI/CD pipelines with blue/green releases, REST, WebSocket and event-driven APIs on Lambda and API Gateway, queues and notifications over SQS and SNS, and large data migrations with Glue and Athena.

On **BuDDy**, a web app for Cucumber test results that I designed and led: live dashboards and PDF and HTML reports, step suggestions and flaky-test analysis with OpenAI's models, a Monaco-based Gherkin editor, and Jira plugins that link tests to issues. Angular, Express, Java, PostgreSQL and Docker.

Web and mobile developer, Foliatech

Tangier · May 2019 to November 2022

SaaS apps across Foliatech's brands, where I looked after the front end in Angular and Ionic, from how it works to how it looks.

Freelance software engineer

Tangier · June 2018 to now

Under the name EOussama, full time at first and alongside my jobs since. The largest is ASIS, a strategic intelligence platform for IRES, Morocco's institute for strategic studies, where analysts get cited answers and briefs from their sources in Arabic, French and English. Others include the websites of Georize, a ground-penetrating radar company, and Samadisa, a construction company, and landing pages, NFT marketplaces and a card battle game for Crypto Katana, Farm Planet, Cybertown, Yukiments and Card Arena. It began with landing pages, forums, stock management systems and online shops.

Game developer, TeamDRD

Tangier · July 2017 to June 2018

Lead developer for a gaming community: I kept up and grew its game mode, written in PAWN and C++.

Education

PhD in artificial intelligence applied to pathological anatomy

Université Hassan II de Casablanca, ENSET Mohammedia · September 2025 to now

My thesis is on the adaptive orchestration and coordination of agentic AI systems in dynamic environments.

Master's in data engineering, Faculté des Sciences de Rabat

October 2023 to June 2025

My final-year project was Patho Draft, report drafting for the pathology department of CHU Mohammed VI in Tangier.

Bachelor's in big data, EST Salé

October 2022 to June 2023

My final-year project was True Drive, which turns any file into a video and back, so that any video host can store files.

Specialised technician in software development, ISTA NTIC Tangier

September 2016 to July 2018

WorldSkills candidate, OFPPT

Marrakech · November 2018 to April 2019

Second in Morocco's national selection, among hundreds of entrants from schools across the country, then trained with five others toward WorldSkills Kazan 2019.

Languages

- **Arabic**, my mother tongue, and Moroccan Darija
- **English**, fluent (C2 reading, writing and listening, C1 speaking)
- **French**, independent (C1 listening, B2 reading, B1 speaking and writing)
- **Italian**, independent (past B1)
- **Spanish**, basic (A2)
- **Japanese**, beginning (A1)

Skills

- **Languages:** Java, TypeScript and JavaScript, C#, Python, Go, Kotlin, Dart, PHP, C and C++, Pawn, SQL, HTML and CSS, shell
- **Front end:** Angular, RxJS and Angular Material, React, Redux and TanStack, Next.js, Vue, Pinia and Nuxt, Svelte and SvelteKit, Tailwind CSS, Sass, shadcn/ui, D3 and Chart.js, GSAP
- **Back end:** Spring Boot, ASP.NET Core and .NET, NestJS, Express, Fastify, tRPC, REST and GraphQL, microservices, JasperReports
- **Mobile, desktop and extensions:** Ionic and Capacitor, Flutter, WinForms, browser extensions with WXT
- **Data:** PostgreSQL and pgvector, MySQL, MongoDB, Redis, DynamoDB, Elasticsearch and OpenSearch, ClickHouse, Prisma and Drizzle, NATS
- **Cloud and delivery:** AWS (CDK, Lambda, API Gateway, SQS and SNS, Glue, Athena, Kinesis), Azure, Firebase, Supabase, Cloudflare, Docker and Podman, GitLab CI/CD, Linux
- **AI:** retrieval-augmented generation, agents, fine-tuning and LoRA, local models with llama.cpp and Whisper, LangChain, Hugging Face, the OpenAI and Groq APIs
- **Quality:** Vitest, Jest, Playwright, accessibility to WCAG 2.1 AA
- **Design:** Figma, Adobe XD, Photoshop, Illustrator

